LOCAL WATER SUPPLIES 17

The **catchment** area is the area from which rainfall and water flow is collected for use by a community.

1 Describe (or provide a map of) your local catchment area.

2 Name some organisations that you could contact to find out more information about your local water supply.

3 Suggest reasons why catchment areas are important.

Contamination occurs when foreign matter gets into water supplies. This lowers the quality of the water and, in severe cases, can make the water unfit for human use.

4 Outline some possible types of contamination in your local catchment area and the sources of these contaminants.

5 There are many chemical tests available for identifying what contaminants are present and the amounts of these substances. Study the list below and outline how each could be identified and measured.

- a Heavy metals
- b Common ions (Name two examples.)
- c Acidity
- d Hardness

6 Suggest why some contaminants can be dangerous in small amounts while others only cause problems in large amounts.

Water is treated before it is supplied to the mains. Each water authority has its own treatment procedure dependent upon the initial quality of water, but the following procedure is fairly typical.

Water from a dam passes through large mesh filters to trap any large objects. The water is clarified (made clearer) by precipitating out some impurities. Alum and iron (III) hydroxide are used as precipitating (flocculating) agents. The larger suspended particles (flocculate) are filtered out, often by passing through a bed of sand. The water is then aerated by cascading through waterfalls or in artificial fountains. This removes odours and tastes by oxidising them. The water can be softened (calcium and magnesium ions removed) by the addition of lime (CaO) and soda (Na_2CO_3). At this stage there may still be microorganisms present so these have to be killed. Chlorine is added (around 1 to 2 ppm) to sterilise the water. Some very resistant microrganisms may survive this process. In some cases ozone (O_3) is used instead of chlorine. Ozone decomposes to form oxygen and is regarded as less of an environmental concern than chlorine. Many areas also add fluoride, which helps protect people's teeth from decay.

7 Define *the mains* in the context of water.

8 Suggest how a bed of sand could be useful in filtering.

9 Draw a flow chart to show the sequence of steps in a typical purification process.

10 The name *alum* can refer to many different chemicals. One alum used in water treatment has the formula $AlK(SO_4)_2.12H_2O$. Name this compound.

11 Give the formula for ozone.

12 Explain why ozone is used in preference to chlorine.

13 Why is fluoride added to water supplies?

14 Give the full chemical name for lime.

15 Give the full chemical name for soda.

16 Read the following passage and define each word in *italics*.

> Microscopic *membrane* filters are used to *eliminate* almost all small *pollutants* in water. They are composed of artificial *polymers* (often polycarbonate), usually consisting of fibres the size of a human hair. The fibres have microscopic holes to filter out impurities, allowing only water to pass through. The holes are smaller than one *micron* (10^{-6} metres) and compared to other processes this process provides improved protection from water-borne *pathogens* such as Giardia and Cryptosporidium.

ISBN 978 1 92561 066 6
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